

MIAE 221: Materials Science for Engineers

Part 2: Chemical Bonding, Potential Energy Wells & Physical Properties

1. The Three Primary (Strong) Chemical Bonds

Primary bonds are strong interatomic attachments involving valence electron transfers or sharing, with bond energies typically ranging from 100 to 1000 kJ/mol (1 to 10 eV/atom).

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[ PRIMARY BONDS ]
|-- IONIC BONDING: Complete transfer of electrons (Cation + Anion)
|                  Non-directional, high melting point, brittle.
|
|-- COVALENT BONDING: Localized sharing of valence electrons
|                  Highly directional, wide range of stiffness.
|
|-- METALLIC BONDING: Positive ion cores in a delocalized "sea of electrons"
|                  Non-directional, ductile, highly conductive.
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A. Ionic Bonding (Electrostatic Attraction)

- **Physical Mechanism:** Occurs between atoms with large differences in electronegativity ($\Delta X > 1.7\text{--}2.0$), typically a metallic element (electropositive, low ionization energy) and a nonmetallic element (electronegative, high electron affinity).



Example: Sodium ($Na : [Ne]3s^1$) transfers its outer electron to Chlorine ($Cl : [Ne]3s^23p^5$), producing mutually stable octets: $Na^+([Ne])$ and $Cl^-([Ar])$.

- **Directionality: Non-directional.** Because an electric field radiates symmetrically in all spherical directions, a positive ion attracts negative ions equally from every orientation in 3D space. The crystal structure is governed purely by ion size ratios (radius ratio r_c/r_a) and charge neutrality.
- **Coulombic Attractive Force:**

$$F_A = \frac{|z_1||z_2|e^2}{4\pi\epsilon_0 r^2}$$

Where z_1, z_2 are ionic valences, $e = 1.602 \times 10^{-19}$ C, $\epsilon_0 = 8.854 \times 10^{-12}$ F/m (permittivity of vacuum), and r is interionic separation.

B. Covalent Bonding (Electron Sharing)

- **Physical Mechanism:** Occurs between atoms with small differences in electronegativity located near each other on the right side of the periodic table (e.g., $C, Si, Ge, O_2, N_2, CH_4$, and polymer chains). Rather than giving up electrons, adjacent atoms share pairs of valence electrons such that each participating atom achieves a stable, closed-shell inert gas configuration.

- **Directionality: Highly Directional.** Covalent bonds form strictly along specific angles defined by quantum orbital overlap (e.g., sp^3 hybrid orbitals form a strict 109.5° tetrahedral geometry in diamond and silicon).
- **Macroscopic Impact:** Because atoms cannot shift without breaking rigid directional orbital bonds, covalent crystals like diamond, silicon carbide (SiC), and silicon nitride (Si_3N_4) possess **enormous hardness, high melting temperatures**, but **no plastic slip**.

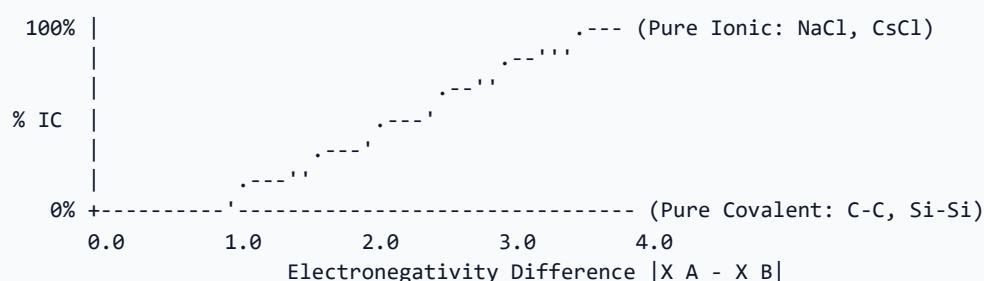
C. Metallic Bonding (The Electron Cloud)

- **Physical Mechanism:** Occurs in pure metals and metallic alloys (Fe, Cu, Al, Ti , brass, bronze). Metallic elements have few valence electrons loosely bound to the nucleus. In a solid, these valence electrons detach from individual atoms and become completely delocalized, forming a mobile "sea of electrons" (or Fermi gas) that freely permeates the entire crystal. The positively charged ion cores (nucleus + inner core electrons) are held together by mutual electrostatic attraction to this permeating electron cloud.
- **Directionality: Non-directional.** The sea of electrons shields adjacent ion cores equally in all directions.
- **Engineering Properties Derived from Metallic Bonding:**
 1. **High Electrical & Thermal Conductivity:** The free valence electrons move instantly in response to electric fields or thermal gradients.
 2. **High Ductility & Formability:** When a shear stress is applied, crystal planes slide over one another (dislocation slip). Because the bonding is non-directional, the electron glue simply flows with the moving atoms—the crystal deforms plastically without cleaving.
 3. **Opaque & Lustrous:** Free electrons absorb and re-emit all visible photon wavelengths.

2. Mixed Bonding & Pauling's Percent Ionic Character

In reality, pure ionic and pure covalent bonds represent idealized extremes. Most chemical compounds exhibit **mixed bonding** characterized by partial ionic and partial covalent character.

Linus Pauling established an empirical equation relating the **Percent Ionic Character (%IC)** of an interatomic bond to the difference in Pauling electronegativities (X_A and X_B):



Reference Pauling Electronegativities

Element	H	C	N	O	F	Na	Mg	Al	Si	Cl	K	Ti	Fe	Zn	Br	Te
X	2.1	2.5	3.0	3.5	4.0	0.9	1.2	1.5	1.8	3.0	0.8	1.5	1.8	1.6	2.8	2.1

Step-by-Step Calculation Examples

1. Titanium Dioxide (TiO_2):

- Electronegativities: $X_O = 3.5$, $X_{Ti} = 1.5 \implies \Delta X = 3.5 - 1.5 = 2.0$

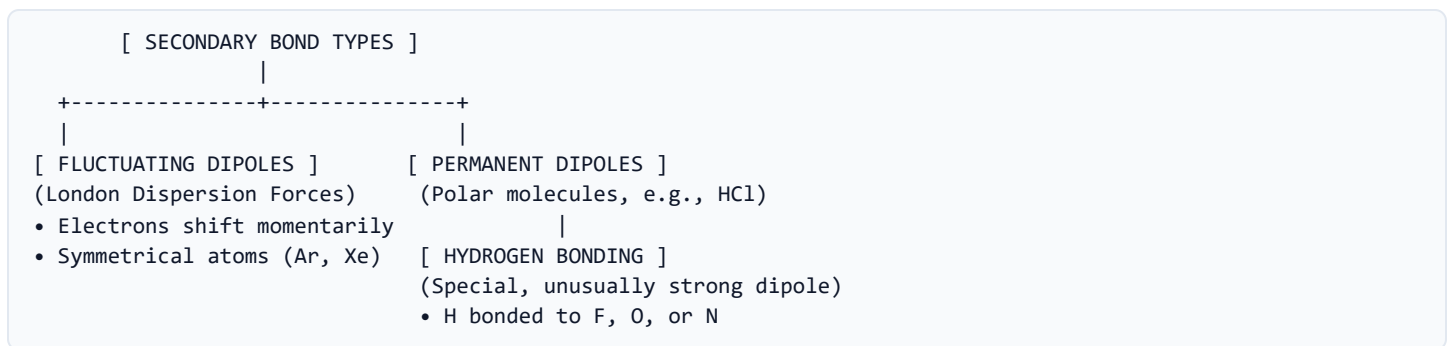
- $(\Delta X)^2 = 2.0^2 = 4.0$
-
- (and 36.8 Covalent).

2. Zinc Telluride ($ZnTe$):

- Electronegativities: $X_{Te} = 2.1$, $X_{Zn} = 1.6 \implies \Delta X = 2.1 - 1.6 = 0.5$
- $(\Delta X)^2 = 0.25$
-
- (and 93.9 Covalent).

3. Secondary (Weak) Intermolecular Bonds

Secondary bonds (often termed **van der Waals forces**) arise from electrostatic attraction between electric dipoles. They carry bond energies between 4 and 40 kJ/mol (0.04 to 0.4 eV/atom)—roughly an order of magnitude weaker than primary bonds.

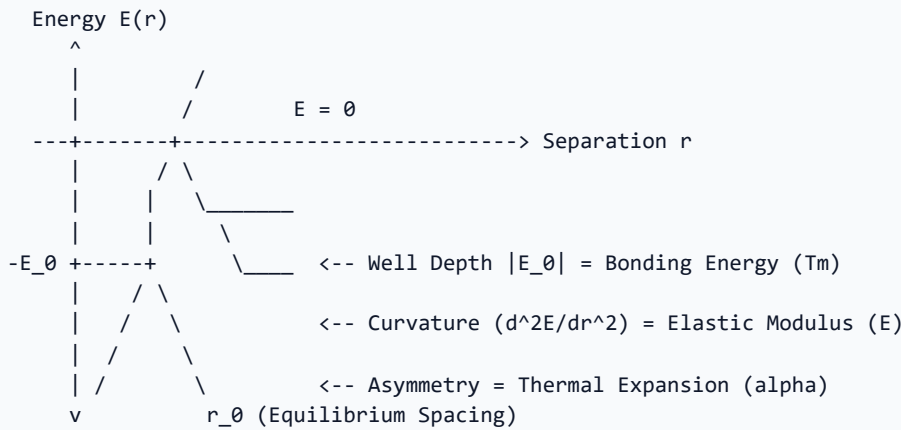


- 1. Fluctuating Induced Dipoles (London Dispersion Forces):** Even in completely symmetric, nonpolar atoms (such as liquid Argon, liquid Helium, or solid Xenon), electrons constantly circulate. At any given instant, random fluctuations shift the electron cloud slightly off-center, creating a fleeting instantaneous dipole ($\delta^+ \dots \delta^-$). This induces a complementary dipole in the neighboring atom, creating a weak attractive force.
- 2. Permanent Dipoles:** Molecules with asymmetric charge distributions (polar molecules, e.g., HCl) possess permanent dipole moments. The positive end of one molecule attracts the negative end of its neighbor.
- 3. Hydrogen Bonding (The Strongest Secondary Bond):** Occurs specifically when hydrogen is covalently bonded to small, aggressively electronegative atoms: **Fluorine (F)**, **Oxygen (O)**, or **Nitrogen (N)**. Because the electronegative atom pulls the shared electron density so intensely toward itself, the hydrogen atom is stripped down to an almost bare proton (δ^+). This exposed positive charge exerts a uniquely strong attraction toward lone electron pairs on adjacent molecules. *Engineering Significance:*
 - Causes water's anomalously high boiling point ($100^\circ C$) and expansion upon freezing.
 - Holds the two strands of the DNA double helix together.
 - Binds adjacent polymer chains in **Nylon** and **Kevlar**, endowing them with extraordinary tensile strength.

4. The Potential Energy Well: The Rosetta Stone of Physical Properties

The shape of the net interatomic potential energy curve $E_N(r)$ directly dictates four major macroscopic engineering properties:

POTENTIAL WELL ANATOMY

Property 1: Melting Temperature (T_m) $\longleftrightarrow$ Well Depth (E_0)

- **Physical Logic:** To melt a solid, thermal energy ($k_B T$) must overcome the interatomic bonding forces holding atoms locked in their lattice sites.
- **Rule:** The deeper the potential well ($|E_0|$ is large), the more thermal energy required to liberate the atoms:

$$T_m \propto |E_0|$$

• **Examples:**

- Tungsten (W): Very deep well ($E_0 \approx 850$ kJ/mol) $\implies T_m = 3422^\circ\text{C}$.
- Aluminum (Al): Moderate well ($E_0 \approx 330$ kJ/mol) $\implies T_m = 660^\circ\text{C}$.
- Mercury (Hg): Shallow well ($E_0 \approx 65$ kJ/mol) $\implies T_m = -39^\circ\text{C}$ (liquid at room temperature).

Property 2: Stiffness & Elastic Modulus (E) $\longleftrightarrow$ Well Curvature

- **Physical Logic:** When an engineer loads a beam in tension, the external stress physically pulls atoms slightly away from their equilibrium spacing (r_0). The material's resistance to this displacement is its **atomic spring constant** (k_0).
- **Derivation:** Near equilibrium ($r \approx r_0$), the net restoring force is approximately linear: $F \approx -k_0 (r - r_0)$ $\implies k_0 = \left. \frac{dF}{dr} \right|_{r=r_0}$ Since $F = -\frac{dE_N}{dr}$, the spring constant is directly the second derivative of the potential energy curve: $k_0 = \left. \frac{d^2 E_N}{dr^2} \right|_{r=r_0}$
- **Rule:** Young's Modulus E is directly proportional to the **curvature (steepness of the bottom)** of the potential well:

$$E \propto \left. \frac{d^2 E_N}{dr^2} \right|_{r=r_0}$$

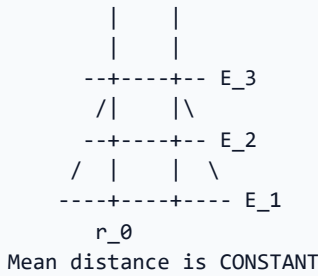
- A **steep, sharp, narrow well** (like diamond or ceramics) resists separation fiercely $\implies$ **High Elastic Modulus** ($E = 300 - 1000$ GPa).
- A **broad, shallow well** (like lead or polymers) separates easily $\implies$ **Low Elastic Modulus** ($E = 1 - 20$ GPa).

Property 3: Thermal Expansion Coefficient (α_l) $\longleftrightarrow$ Well Asymmetry

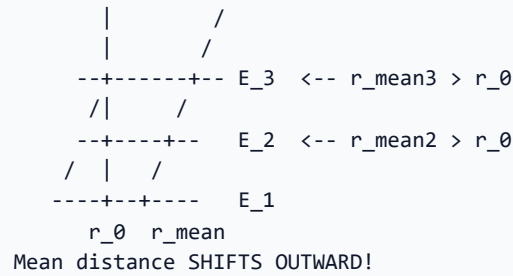
Why do materials expand when heated? Most textbooks state: "atoms vibrate faster." But that alone does not explain expansion! If the potential well were a **perfectly symmetric parabola**, heating would cause atoms to vibrate

symmetrically about r_0 , and the average interatomic distance would **never change** ($\alpha_l = 0$)!

SYMMETRIC WELL
(Hypothetical: $\alpha = 0$)



ASYMMETRIC (REAL) WELL
(Real: $\alpha > 0$)



• **Physical Mechanism:**

1. The repulsive side of the curve ($E_R \propto 1/r^n$) is exceptionally steep because electron clouds cannot interpenetrate.
2. The attractive side ($E_A \propto -1/r$) is much gentler and slopes off gradually.
3. Consequently, the potential well is **strongly asymmetric (anharmonic)**.
4. As temperature rises, atoms gain vibrational energy ($E_1 \rightarrow E_2 \rightarrow E_3$). Because the curve is wider on the right than on the left, the **midpoint of the vibration shifts outward** to a larger average interatomic separation (r_{mean}).

• **Rule:**

- Deep, steep wells (strong bonds, high E_0) are more symmetric $\implies$ **Low thermal expansion coefficient (α_l is small, e.g. fused silica $\alpha \approx 0.5 \times 10^{-6} / \text{K}$).**
- Shallow, highly skewed wells (weak bonds, low E_0) are heavily asymmetric $\implies$ **High thermal expansion coefficient (α_l is large, e.g. polymers $\alpha \approx 100 \times 10^{-6} / \text{K}$).**

$$\alpha_l \propto \frac{1}{|E_0|}$$

5. Master Property-Bonding Correlation Table

Material	Dominant Bonding	Well Depth $ E_0 $	Well Curvature	Well Asymmetry	Melting Point (T_m)	Modulus (E)	Thermal Expansion (α_l)
Diamond (C)	Pure Covalent	Enormous	Very Steep	Very Low	$> 3550^\circ\text{C}$	1050 GPa	$1.2 \times 10^{-6} / \text{K}$
Alumina (Al_2O_3)	Ionic / Covalent	Very Large	Steep	Low	2072°C	390 GPa	$8.0 \times 10^{-6} / \text{K}$
Tungsten (W)	Strong Metallic	Very Large	Steep	Low	3422°C	410 GPa	$4.5 \times 10^{-6} / \text{K}$
Aluminum (Al)	Metallic	Moderate	Medium	Medium	660°C	69 GPa	$23.1 \times 10^{-6} / \text{K}$
Polyethylene (PE)	Secondary (interchain)	Very Small	Shallow	Very High	135°C	1.0 GPa	$200 \times 10^{-6} / \text{K}$